

2(a)	I ref. to features that are absent e.g. no cellular structure any one from nucleic acid core ; either DNA <u>or</u> RNA ; I ref. to, single / double, stranded protein coat / capsid ; A idea of (outer covering) made of <u>capsomeres</u>	1
2(b)	(smallpox) Variola ; (measles) Morbillivirus ;	2
2(c)	allow, virus / viruses, for Morbillivirus I penicillin doesn't act on viruses any two from 1 penicillin only acts on bacteria (and measles is caused by a virus) ; A prokaryotes 2 Morbillivirus does not have, cell walls / murein / peptidoglycan ; 3 Morbillivirus does not have, transpeptidases / the enzyme that is inhibited (by penicillin) ; A idea of not possessing the enzyme that is acted upon by penicillin 4 Morbillivirus does not grow / penicillin (only) acts on growing cells / AW ; A when cell wall is, growing (larger) / getting bigger 5 Morbillivirus does not have cellular structure / Morbillivirus is acellular / penicillin only acts on cells ;	2
2(d)	to treat / to cure / may have / prone to / to prevent / AW, infectious diseases / infections / AW ; e.g. bacterial / opportunistic / secondary, disease(s) / infection(s) A to kill bacterial pathogens A to treat (some) fungal diseases R viral disease / HIV infection do not award if a mix of viral and bacterial diseases stated	1

Question	Answer	Marks
2(e)(i)	<i>allow ref. to, cheaper / low cost, related to relevant mp once only</i> <i>any four from</i> <i>did not mutate to max 2</i> 1 same vaccine could be used (everywhere / for many years) ; A no need to manufacture different vaccines <i>in context of ease of, administering / planning / production</i> 2 (so) a person was protected for, a long time / life / each exposure (to virus) ; AW e.g. same virus so vaccinated person did not become ill on repeated exposure A reduced / no, risk of vaccine becoming ineffective A long-lasting immunity <i>once only for mp2 or mp4</i> I more effective vaccine 3 no, research / trials / development, required for new vaccine(s) ; <i>in context of pre-mass production</i> <i>live and closely related to max 2</i> 4 (closely related so) same antigens / (gives the desired) immune response / antibodies produced / memory cells produced / memory cells remain ; A long-lasting immunity <i>once only for mp4 or mp2</i> 5 <u>live</u> virus (replicates, so), gives a strong(er) immune response ; A better immunity than inactive virus R natural active immunity 6 no need for boosters ; 7 <i>ref. to</i> not smallpox virus, so fewer health and safety issues (at all stages of development and use) ; e.g. could not get ill from smallpox A not smallpox virus so more people willing to have vaccine	4

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2(e)(i)	<i>thermostable vaccine</i> 8 longer shelf-life / doesn't degrade easily / other advantage ; e.g. no refrigeration needed could be easily, stored / transported no wastage <i>easy to administer</i> 9 (using non-medical staff) increased number of people vaccinated / percentage cover / herd immunity ; <i>allow idea of faster delivery of vaccination programme</i> A idea that (makes vaccine) more accessible to people in developing countries											
2(e)(ii)	<i>one mark for correct completion of table ;</i> <table><tr><th>type of immunity</th><th>gained (✓) not gained (x)</th></tr><tr><td>active immunity</td><td>✓</td></tr><tr><td>artificial immunity</td><td>✓</td></tr><tr><td>natural immunity</td><td>x</td></tr><tr><td>passive immunity</td><td>x</td></tr></table>	type of immunity	gained (✓) not gained (x)	active immunity	✓	artificial immunity	✓	natural immunity	x	passive immunity	x	1
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